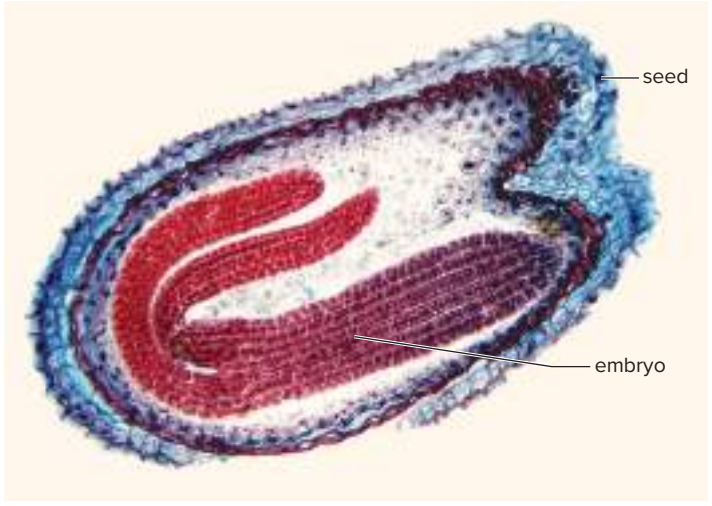


a. Embryo of the green alga, *Oedogonium* 100×



b. Embryo from the land plant, *Capsella* 60×

Figure 23.2 Embryo difference in green algae and land plants. **a.** The filamentous green alga, *Oedogonium,* with an unprotected embryo. **b.** A seed from the shepherd's purse plant, *Capsella,* showing the embryo protected and nourished. (a): Biology Pics/Science Source; (b): Garry DeLong/Science Source



Chara



Coleochaete

Figure 23.3 Charophytes. The charophytes (represented here by *Chara* and *Coleochaete*) are the green algae most closely related to the land plants. (*Chara*): Natural Visions/Alamy Stock Photo; (*Coleochaete*): Dr. Charles F. Delwiche, University of Maryland

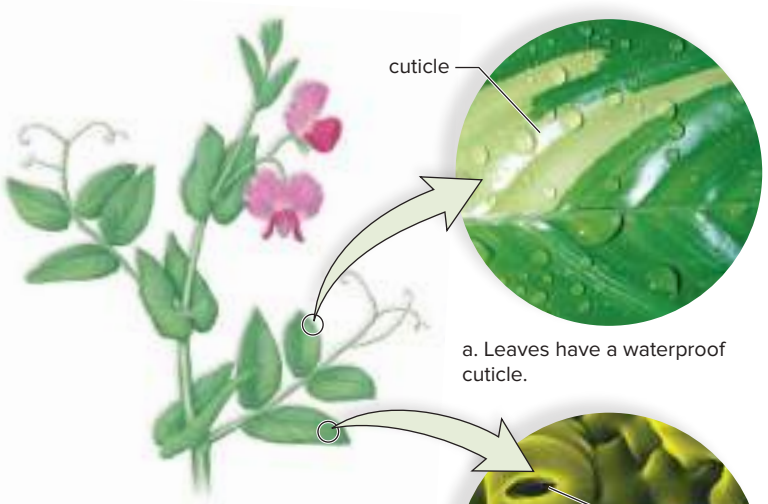


Figure 23.4 Leaf adaptation to prevent desiccation. **a.** A cuticle keeps the underlying cells and tissues from drying out. **b.** The exchange of gases is possible because the cuticle is interrupted by stomata. (a): Ingram Publishing/age fotostock; (b): Dr Jeremy Burgess/Science Source

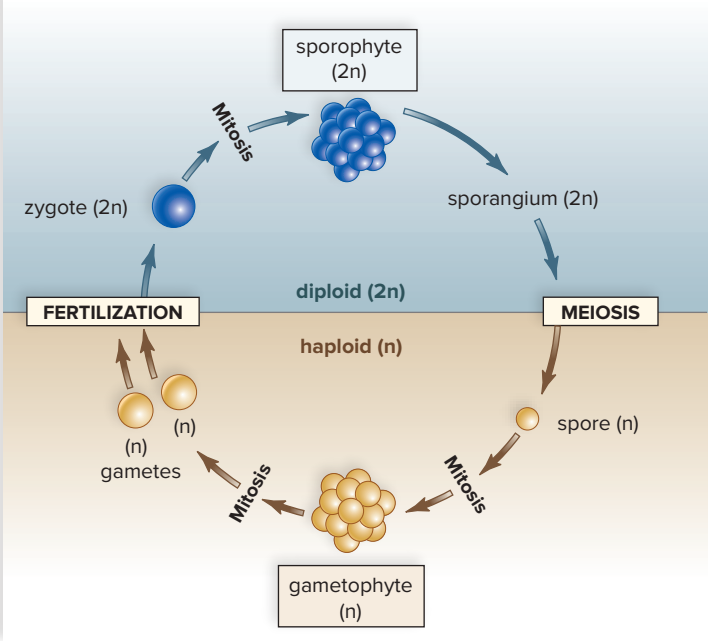


Figure 23.5 Alternation of generations in land plants. The zygote develops into a multicellular 2n generation, and meiosis produces spores in multicellular sporangia. The gametophyte generation produces gametes within multicellular gametangia.

dominant, but in plants such as ferns, pine trees, and peach trees, the sporophyte is dominant (Fig. 23.6). In the history of land plants, a vascular system evolves only in the sporophyte; therefore, the shift to sporophyte dominance is an adaptation to life on land.

Notice that as the sporophyte gains in dominance, the gametophyte becomes microscopic. The gametophyte also becomes dependent on the sporophyte.

ISTUDY